

Video Transcript

Video 3: Basics of Flask Web Server - Part 2 (07:29)

So, we saw last time that we could construct a template using Jinja and here I've got the template called index.html and we put a welcome. But, here we put a variable username and we had this double curly brackets and what this does, it allows us to mix Python with HTML and so, here we've got an if statement and then an end if. So, you'll see statements like this that have one curly brace and then a percentage sign, and it ends in a percentage sign and a curly brace and then we also need an end if.

Let's take a look here at where we pass in this username to this index.html template. So, here in our app, we're passing in the variable called username and here we're setting username to be equal to John. We could change it. We don't have to have username, we can choose any name we want, but then it must match up because that variable is passed into this template and we can pick it up inside the template. So, that's one of the main attributes of Jinja, is that we can pass variables in and data into the HTML. Here also, we can put a placeholder. This one, we start it with block and end it with end block and then we give it a name.

Here, we're calling it content. Later on we can override what's in this block. At the moment, there's nothing in the block, but later on we can override it with what we called a child template and we'll see that later. So, let's now make this a little bit more realistic. I'm going to take the username and I'm going to put it outside here. So, now I'm defining it up here and then I'll pass it in to the template here. Now, let's also have another data structure. Here, I'm defining books and I'm defining a list, and each book is a dictionary.

So key value pairs. Now, what I want to do is I want to render those books. Let me first do it a simple way, and then we'll do it a slightly more complicated way. So, let me return the books to render it. I'm going to use jsonify and I'm just going to pass in books. We imported jsonify here. Okay, so we have access to it, and that's going to take that data structure and it's going to render it. Let's run this. Let's run and debug it, pick up the Python file, and now let's go to local host.

So, this still works, but now I've got another route called books and you'll see that we render at that data structure. Now, let's define ourselves another template to make this more readable. So let's go back to our code and what I want to do now is I'm going to create another template and this template, you'll see it looks very different. It doesn't have any HTML, or it has some in HTML in it, but not much. So here, this extends index and it replaces the block content in index with this. Let's go to index and find where the block content is. It's here, and so we're inserting the books here. We'll still have this printing out, 'Welcome.' Let's just take now a look at what books needs. Well, we've got a for loop in here for book in books. So, we need to pass books in now to this template. Let's see how we do that. So, here we need to take advantage of a Flask function called render template and in that we put the name of the template we want to render. So, the name of this template is books.html and now we need to pass in books and we need to pass in as well, remember the username just like we did here.

So, we'll pass in username equals username. Let's save that and let's go back now and take a look. We need to run it first, so and now let's go back and take a look at books. So, I'm going to reload it. 'Welcome John', and here's the books. So, that's quite neat. We passed in John and we passed in the books and now books inherited index.html and rendered it. So, what happened is that Flask calls the template or rather passes data into Jinja, so our username for example, or the books, and then pulls in the template and populates the template with that data. Here, we were passing in books that inherited index and then that page is passed as the response to the browser. So, that's very neat. Okay, so why don't you now alter the username so that you make it your name, so that you can render it as I've done here so it'll say welcome whatever your name is. Okay, bye for now.